1. Aspect 5: Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation

- **Analysis**: The question directly presents four candidate graphs (A, B, C, D), each attempting to represent the payment schemes of the two newspapers (typically the relationship between sales volume and earnings). Students must convert the textual/visual payment rules from the posters into mathematical functions (e.g., linear, piecewise linear) and then compare them with the graphical representations to determine which graph correctly reflects the two different payment plans. This is essentially **comparing multiple mathematical representations against the same real-world situation**, which is the core of Aspect 5.

2. Aspect 9: Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution

- **Analysis**: Students must clearly understand how the real-world context of "newspaper seller compensation" maps onto the graph axes (sales volume on the horizontal axis, earnings on the vertical axis) and the shape of the curves (slope representing commission rate, intercept representing base salary). For example, if a newspaper offers a fixed base salary, the graph should have a positive earnings value at zero sales; if no base salary, the graph passes through the origin. Students use this context-representation relationship to evaluate the reasonableness of each graph, eliminating options that violate common sense (e.g., earnings decreasing as sales increase). Thus Aspect 9 is highly relevant.

3. Aspect 2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

- **Analysis**: After selecting the correct graph, students need to confirm whether the mathematical relationships described (e.g., a straight line or a broken line) are reasonable in the real context. For instance, earnings should be non-decreasing (usually increasing) with sales, and negative earnings should not appear. Moreover, the two newspapers may have different payment structures (one with high base salary but low commission, the other with no base salary but high commission), which would be represented by different intercepts and slopes on the graph. Students must judge which graphs "make sense" semantically. This directly corresponds to Aspect 2.

4. Aspect 1: Interpret a mathematical result back into the real world context

- **Analysis**: When a student selects a particular graph (e.g., Graph C), they are essentially saying: "In this graph, the line for Newspaper A represents weekly earnings = base salary + commission × sales, and the line for Newspaper B represents ..." Students need to **translate** mathematical features such as points, intercepts, and slopes on the graph back into real-world compensation rules and verify consistency with the posters. This is the process of going from a mathematical representation back to the real situation, aligning with Aspect 1.

5. Aspect 7: Construct and communicate explanations and arguments in the context of the problem

- **Analysis**: Although the question only asks to "circle the correct answer," in the teaching or reasoning process, students need to justify why one graph is correct and the others are wrong. For example, they might argue: "Because the Zedland Star has a base salary, its line should have a positive y-intercept; the Zedland Daily has no base salary, so its line passes through the origin. Only Graph B satisfies this condition." The process of constructing reasons and eliminating incorrect options reflects Aspect 7.

6. Aspect 8: Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions

- **Analysis**: Graphs typically display the payment scheme only within a certain range of sales volumes. Students need to recognise that if the payment rule is piecewise (e.g., higher commission after exceeding a certain sales threshold), the graph should be a broken line, whereas a linear graph would represent a constant commission rate. If the poster rules are linear, the graph should be a straight line; if tiered, the graph should be a broken line. Students must understand whether the chosen graph fully and accurately describes the rules and whether there are unshown intervals (e.g., negative sales are meaningless). This involves awareness of the scope of the representation.

7. Aspect 3: Identify the limits of the model used to solve a problem

- **Analysis**: A graph is a model that may simplify reality (e.g., assuming earnings are strictly linear with sales, ignoring taxes, returns, etc.). Students are not necessarily required to explicitly list limitations, but if a graph shows earnings remaining linear at extremely high sales volumes while in reality there might be caps or diminishing returns, students should be aware that the model may not apply under extreme conditions. In this problem, a linear model is generally valid within a reasonable sales range, but theoretical limitations exist.

8. Aspect 6: Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied

- **Analysis**: In this problem, students might need to consider real-world constraints such as "sellers cannot sell negative quantities" or "sales must be integers," but these do not directly affect graph selection. Thus this aspect is relatively weakly correlated.

1. 方面5：以多种格式解释数学结果与情境的关系；比较或评估两种或多种表示与情境的关系

- **分析**：题目直接给出四个候选图形（A、B、C、D），每个图形都试图表示两家报纸的支付方式（通常是销量与收入的关系）。学生需要将海报中的文字/图像支付规则转化为数学函数（如线性、分段线性等），然后与图形表示进行比对，判断哪一个图形正确反映了两种不同的支付方案。这本质上是**比较多个数学表示与同一现实情境的匹配程度**，是方面5的核心体现。

2. 方面9：理解问题背景与数学解的表示之间的关系，利用这种理解来帮助在背景中解释解，并评估解的可行性和可能的局限性

- **分析**：学生必须清楚理解“报纸销售员薪酬”这一现实背景如何映射到图形中的横轴（销量）、纵轴（收入）以及曲线的形状（斜率代表提成率，截距代表底薪）。例如，若某报纸有固定底薪，则图形在销量为零时收入为正；若无底薪，则图形过原点。学生需利用这种背景-表示的关系来评估每个图形是否合理，排除那些违背常识（如销量增加收入反而下降）的选项。因此方面9高度相关。

3. 方面2：识别数学结果或结论是否在问题背景下有意义

- **分析**：学生选出正确图形后，需确认该图形所描述的数学关系（例如一条直线或折线）在现实背景下是否合理。比如，收入应随销量非递减（通常递增），且不应出现负收入。同时，两个报纸的支付方式可能不同（一个底薪高但提成低，另一个无底薪但提成高），图形上会表现为不同的截距和斜率。学生需要判断哪些图形在语义上“说得通”。这直接对应方面2。

4. 方面1：将数学结果解释回现实世界背景

- **分析**：当学生选定一个图形（例如图形C），他实际上是在说：“这个图形中，报纸A的线表示每周收入 = 底薪 + 提成×销量，报纸B的线表示……”。学生需要将图形上的点、截距、斜率等数学特征**翻译**回现实中的薪酬规则，并验证是否与海报一致。这是从数学表示回到现实情境的解释过程，符合方面1。

5. 方面7：构建并沟通问题背景下的解释和论证

- **分析**：虽然题目只要求“圈出正确答案”，但在教学或思维过程中，学生需要推理为什么某个图形正确而其他错误。例如，可以论证：“因为Zedland Star有底薪，所以它的直线在y轴上的截距应大于0；而Zedland Daily无底薪，所以过原点。只有图形B满足这一条件。”这种构建理由、排除错误选项的过程体现了方面7。

6. 方面8：认识数学概念和数学解的范围与局限性

- **分析**：图形通常只展示了某个销量区间内的支付情况。学生需要意识到，如果支付规则是分段函数（如超过一定销量后提成增加），图形可能是折线，而线性图形只能表示固定提成。题目中如果海报规则是线性的，那么图形应是一条直线；如果是阶梯式，则可能是折线。学生需要理解所选择图形是否完整、准确地描述了规则，以及是否有未展示的区间（如负销量无意义）。这属于对表示范围的认知。

7. 方面3：识别用于解决问题的模型的局限性

- **分析**：图形是一种模型，它可能简化了现实（例如假设收入与销量严格线性，忽略税收、退货等）。学生不一定需要明确列出局限性，但若图形显示收入在极高销量下仍保持线性，而现实中可能有上限或递减报酬，则学生应能意识到模型在极端情况下的不适用。本题中，通常线性模型在合理销量范围内有效，但理论上存在局限性。

8. 方面6：使用现实世界如何影响数学过程或模型结果的知识，做出上下文判断

- **分析**：本题中，学生可能需要考虑“报纸销售员不能卖负数量”“销量应为整数”等现实约束，但这些并不直接影响图形选择。相对而言，此方面相关性较弱。